Pamantasan ng Cabuyao 3

Transcribed by TurboScribe. Go Unlimited to remove this message.

Great. Later, bakit monitoring na tayo? Meron tayo dito sa advisors. Sa center dito.

So ngayon, rest muna kayo pero bukas starting tomorrow. And then, hindi mo yung... Pwede kayo... Isa plage yun, yung... You can mute for the safety of everyone. Okay.

Sige, yung window. Ayan. Sige.

Ayan. Yes. Good luck.

Good luck, mga boss. Mabasa yung text. Let's go back.

Kayang kain niya. Ikaw na. Let me introduce you to the panel member.

On my right side, ginagalang orkest, our internet expert. And then, our partner faculty. And on my left side... And then, on my left side, MNEC Tapuno.

Consultant, Data Privacy. And... Parang may nasyang talk. And of course, one of the pioneer computer engineers.

We're back, but the school is now based in Egypt. Kasi na... Pero may akong panigus. Ang pinaka-office, operations build.

Pero meron na rin na mong open na mga subjects. At away ang presidente, at saka vice-president. Gano'n yung darating ang nangyari.

Sana hindi yung... Yung bomba. Ay, grabe. Kasi supported ng Russia, eh.

Ang vice-president. Okay. Ang Saudi, ang nakabaka kay Russia.

Gano'n ba nga? Kaya maluling... Nagibagal dito yung... Yes. Okay, so... Demising... Okay, so before we proceed, first, introduce yourself. So, good morning, greetings, and good morning, members of the panel.

I'm Charlie May Aratas. I'm presenting with my co-members. Oliver po yung... Okay.

Noted. So, as a student here in our beloved university, we've experienced how disconcerting it is to look for library information. Gusto nyo kareklama ka.

So, queuing at the registrar, asking... asking professors, or... walking from office to office is only endowed with unverified or conflicting details. So, and this is just a story. I will hand over the floor to my co-presenter, Mr. Anciso, to walk you through to our next slide.

Nag-survey po kami ng 38 students, and out of 34, we have 69.5%. Eh, nasabi po nila na nahihirapan po silang makahalap namin. Ah, di naman sa matino, pero, ah, conflict information po. So, dahil nga po sa fragmented distribution, matalas, makakuha nila sa friends, or online, or, ayun, sa LMS.

So, this highlights the need for an easily accessible delivery with verified and patient answers directed to students. So, ayun nga po. The 34% are students that, that are yung nagre-reklamo po natin for policies.

The 31.6% are from the scattered or unread documents, which is labahan. Parang mga salito siya sa ating mga documents namin. And, ah, 39% is the need to ask others of this group, ask the other students for the information.

So, our, ah, solution sa problem, ah, we, ah, are going to implement, ah, offline capable, ah, chatbot that can run on small language mobile, that can cite grounded answer from manuals, and can run on a consumer GPU. So, eh, kailangan po siya quantize. And then, kailangan po siya tune-in so we can improve the throughput kasi ma-reduce natin yung system from SNAP by pa-tuning down behaviors and, ah, behavior and, ah, ah, behavior.

For our main objective, ah, we want to develop and evaluate an offline university, ah, an offline university helpdesk that will answer the student, ah, inquiries directly from our official manuals. So, to be practical, to be practical and to be deployable, ah, ah, the entire system will run, ah, using 4-bit quantized Lama 3.2 ah, model with, ah, 8-bit Lama. Um, by doing this, the, the student will get the accurate and grounded answers without needing the internet and expensive EPIs.

So, for our objective one, the first one is the Lama 3.2 model with response time, speed, and memory. So, we are measuring here if ah, the, the, ah, solution is, ah, deployable in the consumer GPU. So, if it can run in the consumer GPU only.

So, we are measuring response time, speed, and memory. Then, we are building a knowledge base from the official manuals. So, we are building a sanitized knowledge base for, ah, the, the unnecessary like, the, the, the names on the top, or the pages, pages that refer to the other pages.

I just removed the unnecessary information. I just included the necessary passages. Then, I just applied to compare with the adapted model of correctness, instruction, following, and support test series.

So, for our objective three, ah, this is where our baseline is that the unadapted model is, ah, compared to the QLARA fine tune for behavioral training. Then, objective four is to implement RAD pipeline. So, for our objective two, this is our hybrid retrieval pipeline.

objective three, So, for our objective three, this is our hybrid retrieval pipeline. So, for our

objective three, this is our So, for our objective three, this is our hybrid retrieval pipeline. So, for our objective three, this is our hybrid retrieval pipeline.

So, for our objective three, this is our hybrid retrieval So, for our objective three, this is our hybrid retrieval So, objective three, this is our So, for our objective three, pipeline. So, for our objective three, this is our pipeline. So, for our objective three, for our objective three, pipeline.

So, for our So, objective three, pipeline. So, for our We will be using a 4-bit LAMP 3.2 on a single machine with a TPU using a hybrid drive over T-models to protect the student privacy also. So we limit that and under, so to protect student privacy under 10.170 which are TPU stability, binalimited personal records and multi-insertion.

So personal records queries route to path B producing 0 fake numbers. So next is the research box. So this is an offline execution on limited hardware.

So number 1 here is lab of real-world testing software. So there is no research that measures latency and generation-specific memory specifically on 8-bit DRAM. So number 2 is reference behavior to QLORA.

So same thing, there is no data that shows how consistent QLORA is training 3D models specifically on S-based university reference. So next is the citation error knowledge base. So this is one of the critical research.

So this is one of the critical research. So this is one of the critical research. So next would be small models for university needs.

So most academic literature evaluates massive expensive cloud-based models. So paramiyan po kasi ng mga researchers po kung anong case na cloud-based po. So since offline po, gumamit po kami ng small models.

So for the system design, this section summarizes our hardware distribution. So we divided it into 3 workloads para hindi po mag-processing po. So first po sa GPU po, we use the RTX 4060.

And then yung only job lang po is to run the 4-bit LAMAM model and then the QLORA adapter. So lahat po ng mga generations or generated na answers po is sa GPU lang po mag-use it. And then next po for our CPU po, the AMD Ryzen 5. Since medyo tight po yung GPU, lahat po ng mga non-LLM tasks like searching, checking citations, doon po lahat we work.

And then lastly po, the storage and the OS. Doon po namin ni-store yung mga model weights and yung mga PDFs. Locally, then i-run po siya through Windows 11 and WSL 2. So this is our entry read by type.

So normally input and then converted into vectors and indexes retrieval. Then gate. So ang gate 1 po namin is, pupunin niya yung top answers nung both retrieval, nung hybrid.

Then i-store niya kung enough by evidence. So pag-enough, path A cited answer yung normal generation po natin. Ibig sabihin may enough evidence siya para bigyan na answer yung student.

Then for path B, ibig sabihin wala po siya ng enough evidence. May evidences po siya pero hindi tayong subutin ng model. Kasi baka personal or super complicated na wala naman sa manual.

Pero may evidence, may office na kailangan po. Tapos pag out of scope na po talaga, may fixed message or static message lang po tayo, doon po natin kailangan po tayong model generation. And then lahat po yan, lahat ng possible paths na lahat po ayan.

So this slide is a proof that our solution is cost effective without relying on too much expensive server infrastructure. So here, we do not feed raw PDF into the AI. So, strict non-procedural front model from three official manuals of the university.

So, chunk text along structural heading at max of 512 copies. So here, in page, an admission game rejects missing field. This enables our citation verifier to catch hallucinations.

So, development process naman po. So yung increment one, before kami magsulat na code or mag-train po natin. So, kailangan po muna naman mag-establish ng physical constraint po ng system.

So, yun na nga po yung hardware, tsaka yung knowledge po naman. Then, biniberal tuning and padline. This is the part where we teach the model and kung paano po siya nag-ilos, kasabay ng pag-integrate po ng search mechanism.

Yung VM25 po, tsaka yung vector search. So, next po is routing and final benchmark. Ito po yung sa increment pipe and post-development.

Then, above naman po po sa optimization tsaka efficiency po nung system namin. Then, dito na po namin po prove kung okay po yung binibigay na subot po naman. For evaluation plan po, dito po namin in-ensure that yung experimental result po namin is mag-stay na completely honest and unbiased po.

So, may lima po kaming set, strictly set. First is yung evaluation set po. Ang evaluation set po is may 80 queries na may 40 answerable lang po.

And then, 25 na yung scope but walang evidence. Then, out of scope po. Ginagamit lang po ito for final eval lang po.

So, hindi po talaga siya pag nasa development part po siya, hindi po siya mag-run po. And then, for development set po, same po ng proportion po sa task. May 40 queries.

And then, ginagamit lang po ito to check the increment threshold po. So, parang ito po yung magiging math test po sa pagdating po sa mga questions po. And then, for calibration set po, we

have 10 queries po.

And, ginagamit lang po ito for response times and speed limits po. And then, for training set po, we have 300 examples across the 5 dataset po. Neutalize niya lang po yung QLORA behavior and fine tuning.

And then, lastly po, we have the scope examples. Ginagamit lang po ito for gate 2. Patulad po nang sinabi kanina na kapag hindi po... ...pag nag-fail po si gate 1, yung retrieval niya po hindi po enough or strong enough po dun sa hinahanap na kailangan ni student, papasok lang po si gate 2. So, ito po yung 10 measures namin, yung mga measure po namin na alam niyo sa objective. So, deployment, ayun nga po yung quick rerun, response time, generation speed, and throughput.

Yan po yung sa objective 1, para malaman po natin kung kaya pa tayo naman sa consumer GPU lang po. Yung sa fine tuning, uncertainty, instruction following, and unsupported queries. Yung mga measure po natin para malaman po natin kung tama yung bibigay ng pipeline po natin.

At saka nung kung tama po yung pagkakatrain po sa QRA adapted. Dun sa pipeline, retrieval accuracy, routing accuracy, malalaman po natin dito kung tama po yung... Out of scope. So, for this slide, it show how we evaluate the direct impact of QRA fine tuning by comparing the base model against the adapted version.

It sets clear target metrics for answer correctness, instruction following, and the model's ability to deline missing data without making observations. So, these presents are pre-registered metrics. So, all our mechanics of set-task benchmark including the vertical, strict correctness, and zero dangerous errors were locked before final testing to eliminate post-hoc human bias.

So, our dangerous error rate is strictly 0 to 15, ensuring zero out of scope queries produced by policy. Next is, dito po namin mapapasesta kung gano'n po ka-transfer. So, first one is separate data sets.

So, no condition we will use to test the transcript of the variant must be completely different from the example we used to predict. So, pag nag-undergo na po kami ng final testing is hindi na po kami gumagamit ng mga pre-questions po. Umbaga is, i-stress test po namin yung system.

Then, next is target peaks before testing. So, yun po, hindi po namin madagaya kasi gawaan na meron po kami, di ba yung, umbaga para po kami may increase. Kung meron po kami, hingap yung target na impossible po na pwede pong ma-reach po namin or hindi.

Then, next is frozen repeatable runs. So, next is freeze everything manuals, chunks, and the AI settings so that there are running the test 10 times with the exact same output every single time. So, para kung, ano siya, basically para siyang T-switch, umbaga.

Tapos, um, para dito pa namin papakita kung talaga yung sinasagot po ng AIP kahit yung

questions po ay medyo malayo na or ano. Hindi na po, wala na po si scope is, ah, nagiging anod pa rin po yung answer nun. Tapos tama, tama din po yung answer nun, ano yung chunk.

Then, next is single runs. So, the automated testing script uses the exact same underlined functions. So, the routing and handling, the graphical type, the student uses.

So, for ISO-IEC, um, 25010 of 2020. Um, dito po, ah, system, ano po, system level software evaluation po siya. So, um, hindi lang po siya puro about sa consumptions and anything.

Um, e-evaluate din po namin yung mismo, ano, yung entire health test po. So, meron po aming three core quality po. Um, the functional suitability.

For, ito po kung working po talaga siya. Ah, as intended. And then, performance efficiency.

Um, dito po maano kung yung, kung gaano po kabilis yung response time niya po. And then for, last is the reliability. Um, kung magiging consistent po ba yung health test po namin.

And then, to validate po yung tatlong objective po namin. We have, um, panel, we have panel na three to five experts. Um, with two technical plus po yung one to three po from the offices na gumagami po po ng manual, manual system po.

Um, like registrar po, mga student affairs po. And then, for, um, para po ma-evaluate po namin. Um, magkakondak po kami ng test ng problem system.

Nagkakonsist po siya ng 30 minute guided demo po using 15 fixed questions po. And then, to, for, the evaluators will assess function. Um, the efficiency by five point likelihood scale po.

That's all. Mag-explore yun po. Ah, na-automate po namin dito yung, ah, yung, yung sakayolara po, yung, ah, kasi po yung pagkina-quantize po tayo ng model po, ah, mga baba pa yung taxation eh.

Pero, yes po, may, meron po tayong lag na may papakain sa model. Pero, yung way niya po ng pagsalot na, pagka-quantize po kasi ng model, based sa, ah, ano yun. Ang in-optimize yun yung flow ng data.

Ah, po. Pero sabi niyo dito yung LAMA. LAMA 3.2. LAMA 3.2. Optimize 3.2 LAMA through 4B quantization and QLORA.

So, ngayon, QLORA yung in-optimize yun. Ngayon, anong optimization technique yung ginawa nito para na-optimize yung QLORA? Ah, muna po yung routing technique po namin. Ah, mas malalasin natin yung hallucination by, ah, routing through, ah, if it says, ah, kung tama yung, ah, kung may enough evidence, magbigay natin siya sa model ng citation.

Then, sa, ah, pagka walang evidence, ah, static message. And then, ano ba? Anong

optimization? Ah, routing. Diba may mga parameters yung QLORA? Yes, po.

Diba tatapina yung in-optimize natin? Kasama ba yung routing doon sa QLORA? Ah, kasi. Parameters. Ang pinaka-in-optimize, ang pinaka-optimize sa training ng QLORA is yung training set, po.

Kasama po din yung sa training set po namin. Ah, kasi hindi malinaw dito. Wala na pala dito.

Kasi, ang in-expected dyan, ah, meron kayong baseline, no? Yes, po. Ang baseline yun, yung LAMA. Yung unadapted, po.

Baseline yung LAMA. Local, no? Tapos, papasokan yun siya ng 4-bit quantization. Yes, po.

Tsaka, ah, pwede yun na-adaptor ako, po. O, yun na yun. So, may...

This file is longer than 30 minutes.

Go Unlimited at https://turboscribe.ai/ to transcribe files up to 10 hours long.